

Structured 3D from Language. Every Gaussian Has a Name.

Recursive semantic-to-geometric decomposition via Gaussian splatting. A language model breaks prompts into composition trees, each leaf becomes a named Gaussian primitive.

May 2026 / Pre-seed

The Problem

Current text-to-3D produces monolithic, non-divisible output

Today's generative approaches predict pixels or radiance fields directly through diffusion. They do not produce structured, decomposed, editable scenes. None use compositional Gaussians. None produce labeled parts. None allow editing without full re-generation.

- Video diffusion (Sora, Runway): non-divisible frame sequences
- Image diffusion (DALL-E, Midjourney): flat rasters, no geometry
- 3D diffusion (DreamFusion, Point-E): opaque radiance fields

Monolithic (Status Quo)

Input: prompt. Output: 10M unstructured floats.
No parts. No labels. Edit = start over entirely.

Structured (SGS)

Input: prompt. Output: Gaussian space, renderable,
decomposed into named blobs. Edit one node.

Our Solution

Recursive Semantic-to-Geometric Decomposition (RSGD)

A language model decomposes prompts into composition trees. Each node is a named sub-concept with spatial relations. Terminal nodes become individual Gaussian splats. The tree IS the editable representation.

```
"a castle on a hill"
|
[Decomposer LM] --> Composition Tree:
|               scene
v               +-- castle
[Recursive      |   +-- tower_1
Renderer]       |   +-- keep
|               +-- hill
v
Editable Scene Graph + 3D Render
```

What Is Semantic Gaussian Splatting?

SGS unifies language understanding and 3D geometry under one equation. A single Gaussian primitive encodes both spatial structure and semantic meaning.

The SGS Primitive: $G = (\mu, \Sigma, \alpha, f)$

- μ (position): center point in splatting space
- Σ (covariance): shape and orientation of the ellipsoid
- α (opacity): transmittance weight during rendering
- f (features): semantic embedding vector encoding meaning

Key Insight

Alpha-compositing is a strict superset of softmax attention. Any transformer layer can be expressed as Gaussian rendering. Formally verified in Lean 4. Submitted to JMLR.

What Are Semantic Gaussian Blobs?

A blob is a pre-computed chunk of meaning stored as a Gaussian. One blob = one passage, concept, or object compressed into a single primitive. Blobs bridge language memory and 3D scene composition.

Two-Pass Rendering

- Pass 1 (blobs): Retrieve background knowledge by cosine similarity over blob embeddings. These form the scene context.
- Pass 2 (words): Alpha-composite the current input over the retrieved blob layer.

This lets the system recall prior scenes, reuse memorized components, and compose new structures without retraining.

How We Use SGS + Blobs to Render

Four-stage pipeline from prompt to final scene

1. Decomposition: LM parses prompt into composition tree
2. Blob Mapping: each node maps to shape distributions from vocabulary
3. Gaussian Instantiation: terminal nodes become $G = (\mu, \Sigma, \alpha, f)$
4. Alpha-Compositing: front-to-back transmittance-weighted rendering

The same mathematical operation appears at every stage.

```
PROMPT
|
[DECOMPOSER LM]
|
[BLOB MAPPING]
|
[INSTANTIATION]
|
[ALPHA-COMPOSE]
|
RENDERED 3D SCENE
```

Key Innovations

Five pillars from theory to implementation

Alpha-Compositing Superset Proof

Strict superset of softmax attention. Lean 4 verified. Submitted to JMLR.

Recursive Decomposition

Prompts decompose into trees, trees into sub-trees, leaves into Gaussians. The hierarchy IS the scene graph.

Conversation Memory via Blobs

Each conversational turn stored as Gaussian blob. 40% retrieval recall at 90-turn distance. Flat cost per turn.

Editable Scene Graphs (DSL v1)

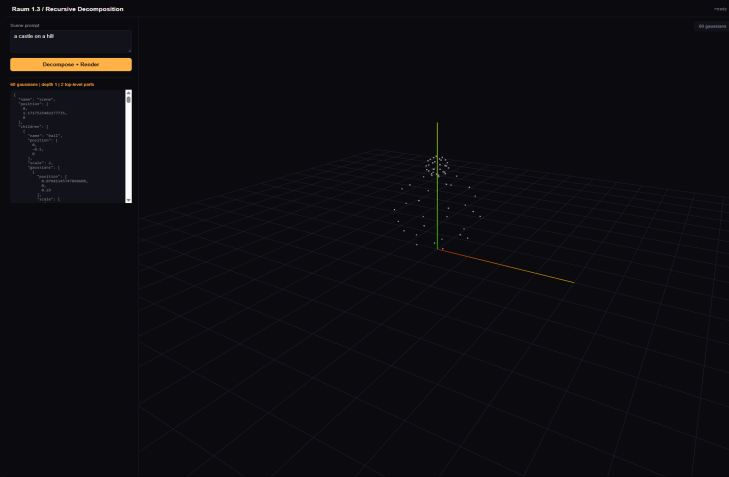
Declarative DSL describes scenes as labeled nodes. Edit by name, reposition, recolor. Re-render is instant.

OOV Policy (Out-of-Vocabulary)

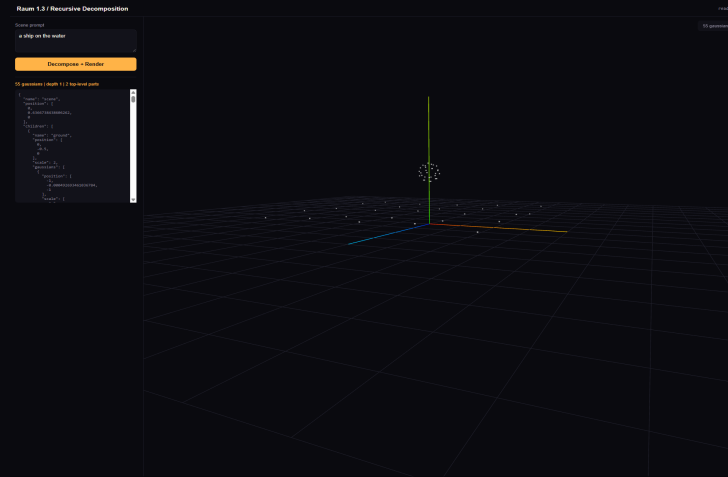
Unknown objects resolved via GloVe cosine nearest-neighbor across 300-class vocabulary.

Demonstration: Model-Generated

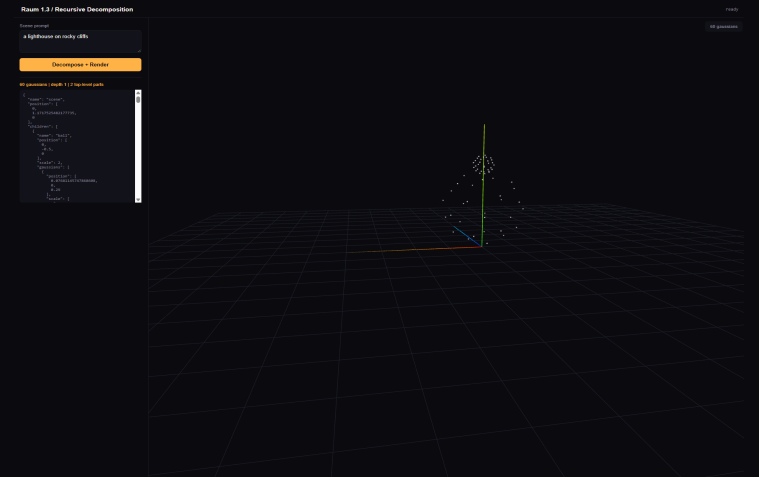
Planck 1.3 (100M params, structure-only generation)



"a castle on a hill"



"a ship on the water"

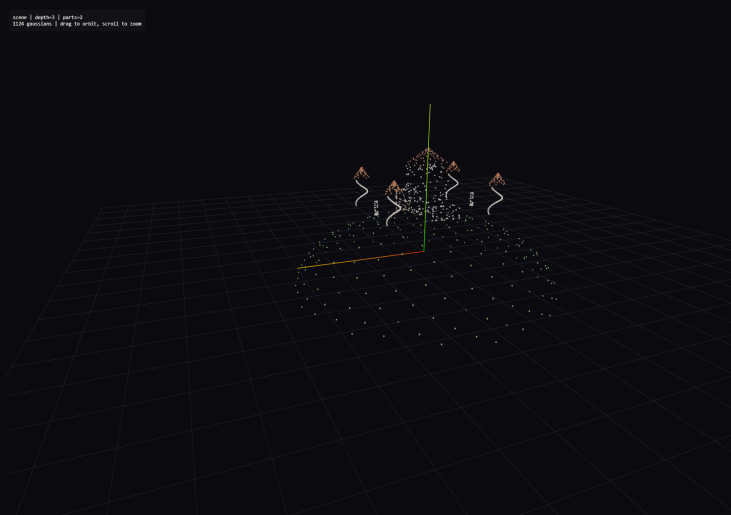


"a lighthouse on rocks"

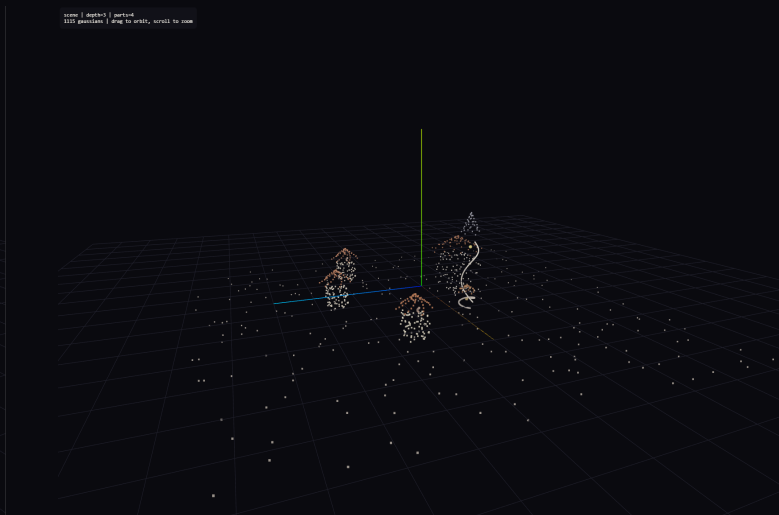
Limited by 512-token context. Hertz (1B+) will produce richer, deeper composition trees.

Iteration 2 with Larger Model

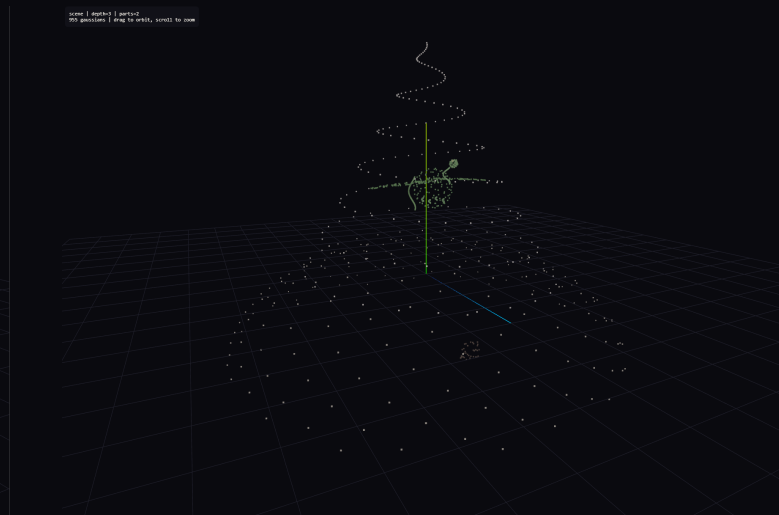
Hand-authored composition trees rendered through the same pipeline. Target quality for Hertz-scale generation.



Castle (1,124 Gaussians)



Village (1,115 Gaussians)



Station (955 Gaussians)

Applications

Structured, queryable, recomposable 3D scenes

- Game Asset Pipelines: editable scenes with labeled buildings, props, terrain. Export parts to engines.
- Architectural Visualization: building descriptions to 3D walkthroughs. Change materials per room.
- Film Pre-visualization: script descriptions to scene layouts. Directors iterate on composition.
- Education and Science: labeled 3D models with queryable parts. Hide/show layers interactively.
- Conversational 3D: retrieve prior scene blobs, modify specific nodes, re-render without regeneration.
- Accessibility: every blob has a semantic label. Screen readers describe spatial relationships.

Roadmap

From proof-of-concept to production API

Completed

- Planck 1.0-1.3: 100M LM, blob retrieval proven
- Raum 0.1: 300-class text-to-3D demo, editable DSL
- Raum 1.3: recursive decomposition pipeline
- JMLR: alpha-compositing superset proof (Lean 4)
- Planck 1.4: conversation memory blobs

Next

- Hertz 1.2 (Q3 2026): 1B param, 2048 context
- Raum 2.0 (Q3 2026): Hertz-powered decomposer
- API Beta (Q4 2026): prompt to structured 3D JSON
- SDK (Q1 2027): Unity and Unreal Engine plugins

Structured 3D. From Language. At Scale.

We are building the decomposition layer between language models and 3D rendering. Every Gaussian has a name, a parent, and a position in a tree that mirrors how humans think about scenes.

Nikita Gorshkov, Founder

Pre-seed funding to scale Decomposer LM (1B), build API infrastructure, hire rendering engineer.

github.com/feamando/sgs